

Benchmark Report

Benchmarking of NQCH’s quantum computer

May 6, 2026

1. Report of Changes

**Platform:** sinq20  
**Calibration-id:** *ec305fcaf377a05f4fb391a8ee18282bfae18df6*  
**Calibration date:** 2026-04-28 07:58:51  
**Calibration note:** chore(sinq20): update readout parameters  
  
**Experiment-id:** 20260504115618  
**Experiment date:** 2026-05-04 11:56:19  
**Experiment note:** *temporary note!!!*

**Platform:** sinq20  
**Calibration-id:** *4dc4082f38a53222b3956c22202d32a520d4bc78*  
**Calibration date:** 2026-01-15 02:03:03  
**Calibration note:** chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9  
  
**Experiment-id:** 20260118161538  
**Experiment date:** 2026-01-18 16:15:38  
**Experiment note:** *temporary note!!!*

2. Version Comparison

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.23  | numpy   | 2.4.4   |
| qibolab      | 0.2.9   | qibocal | 0.2.5   |
| matplotlib   | 3.10.8  | scipy   | 1.17.1  |
| scikit-learn | 1.8.0   | pandas  | 2.3.3   |
| networkx     | 3.6.1   | sympy   | 1.14.0  |
| torch        | 2.11.0  |         |         |

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.22  | numpy   | 2.2.6   |
| qibolab      | 0.2.9   | qibocal | 0.2.4   |
| matplotlib   | 3.10.3  | scipy   | 1.15.3  |
| scikit-learn | 1.6.1   | pandas  | 2.2.3   |
| networkx     | 3.4.2   | sympy   | 1.14.0  |
| torch        | 2.7.0   |         |         |

3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



Q17

Q18

Q19

Q12

Q13

Q14

Q15

Q16

Q7

Q8

Q9

Q10

Q11

Q2

Q3

Q4

Q5

Q6

Q0

Q1

88.65

89.85

91.05

92.25

93.45

94.65

95.85

97.05

98.25

99.45

1Q Fidelity

#### 4. Statistics

|                | Average | Median | Min   | Max   |
|----------------|---------|--------|-------|-------|
| T1 (ns)        | 23.9    | 23.8   | 18.3  | 29.6  |
| T2 (ns)        | 9.84    | 9.66   | 1.52  | 22.2  |
| Fidelity       | 0.975   | 0.968  | 0.94  | 1     |
| RO Fidelity    | 0.971   | 0.971  | 0.964 | 0.981 |
| 2q RB Fidelity | -       | -      | -     | -     |
| 2q CZ Fidelity | -       | -      | -     | -     |

|                | Average  | Median   | Min   | Max      |
|----------------|----------|----------|-------|----------|
| T1 (ns)        | 1.28e+04 | 1.23e+04 | 646   | 3.65e+04 |
| T2 (ns)        | 2.36e+25 | 3.68e+03 | 125   | 9.43e+26 |
| Fidelity       | 0.99     | 0.997    | 0.887 | 0.999    |
| RO Fidelity    | 0.794    | 0.777    | 0.777 | 0.927    |
| 2q RB Fidelity | 0.965    | 0.972    | 0.923 | 0.981    |
| 2q CZ Fidelity | 0.975    | 0.974    | 0.967 | 0.991    |

#### 5. Best Qubits Selection

| k-qubits | Best Qubits    | Fidelity |
|----------|----------------|----------|
| 2        | 0, 3           | 0.909    |
| 3        | 0, 1, 3        | 0.899    |
| 4        | 12, 13, 17, 18 | 0.891    |
| 5        | 0, 1, 3, 4, 8  | 0.886    |

| k-qubits | Best Qubits   | Fidelity |
|----------|---------------|----------|
| 2        | 2, 3          | 0.981    |
| 3        | 0, 2, 3       | 0.976    |
| 4        | 0, 1, 2, 3    | 0.970    |
| 5        | 0, 1, 2, 3, 4 | 0.965    |

#### 6. Summary metrics

| Benchmark       | Metric   | Value  | Runtime    |
|-----------------|----------|--------|------------|
| Mermin          | Max      | -3.246 | 207.94 sec |
| Grover 2q       | Max prob | 0.855  | 3.85 sec   |
| Grover 3q       | Max prob | 0.294  | 2.46 sec   |
| GHZ state prep. | Success  | 0.816  | 2.22 sec   |

| Benchmark       | Metric   | Value  | Runtime  |
|-----------------|----------|--------|----------|
| Mermin          | Max      | -3.316 | 1.02 sec |
| Grover 2q       | Max prob | 0.89   | 2.36 sec |
| Grover 3q       | Max prob | 0.385  | 2.43 sec |
| GHZ state prep. | Success  | 0.910  | 3.95 sec |

#### 7. Randomized Benchmarking Results

| Qubit n | Fidelity | Error Bars | Qubit n | Fidelity | Error Bars |
|---------|----------|------------|---------|----------|------------|
| 0       | 0.995    | ± 0.000224 | 10      | 0.996    | ± 0.00012  |
| 1       | 0.991    | ± 0.000662 | 11      | 0.997    | ± 0.000156 |
| 2       | 0.998    | ± 8.9e-05  | 12      | 0.998    | ± 7.26e-05 |
| 3       | 0.998    | ± 7.97e-05 | 13      | 0.998    | ± 9.89e-05 |
| 4       | 0.991    | ± 0.000559 | 14      | 0.998    | ± 5.64e-05 |
| 5       | 0.997    | ± 0.000197 | 15      | 0.998    | ± 0.000118 |
| 6       | 0.994    | ± 0.00018  | 16      | 0.982    | ± 0.00376  |
| 7       | 0.976    | ± 0.00421  | 17      | 0.956    | ± 0.0121   |
| 8       | 0.998    | ± 6.71e-05 | 18      | 0.995    | ± 0.000286 |
| 9       | 0.998    | ± 8.32e-05 | 19      | 0.996    | ± 0.000122 |

| Qubit n | Fidelity | Error Bars | Qubit n | Fidelity | Error Bars |
|---------|----------|------------|---------|----------|------------|
| 0       | 0.997    | ± 0.000131 | 10      | 0.989    | ± 0.000582 |
| 1       | 0.998    | ± 7.47e-05 | 11      | 0.996    | ± 0.000218 |
| 2       | 0.998    | ± 7.57e-05 | 12      | 0.998    | ± 0.000126 |
| 3       | 0.995    | ± 0.000163 | 13      | 0.998    | ± 7.15e-05 |
| 4       | 0.996    | ± 0.000185 | 14      | 0.985    | ± 0.00183  |
| 5       | 0.995    | ± 0.000147 | 15      | 0.995    | ± 0.000469 |
| 6       | 0.998    | ± 8.34e-05 | 16      | 0.993    | ± 0.000558 |
| 7       | 0.887    | ± 0.0287   | 17      | 0.99     | ± 0.000502 |
| 8       | 0.999    | ± 4.48e-05 | 18      | 0.997    | ± 0.00026  |
| 9       | 0.998    | ± 7.2e-05  | 19      | 0.998    | ± 9.05e-05 |

## 8. Mermin

Mermin's algorithm for 3 qubits.

| Runtime    | Qubits  | Max    |
|------------|---------|--------|
| 207.94 sec | 0, 1, 3 | -3.246 |



| Runtime  | Qubits  | Max    |
|----------|---------|--------|
| 1.02 sec | 0, 2, 3 | -3.316 |



## 9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in  $[[0, 3]]$ .

| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 3.85 sec | 0, 3   | 0.855    |



| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 2.36 sec | 2, 3   | 0.89     |



## 10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in  $[[0, 1], [0, 3]]$ .

| Runtime  | Qubits  | Max prob |
|----------|---------|----------|
| 2.46 sec | 1, 3, 0 | 0.294    |



| Runtime  | Qubits  | Max prob |
|----------|---------|----------|
| 2.43 sec | 0, 2, 3 | 0.385    |



## 11. GHZ state preparation

GHZ circuit with 3 qubits executed on sinq20 backend with 1000 shots.

| Runtime  | Qubits  | Success |
|----------|---------|---------|
| 2.22 sec | 1, 0, 3 | 0.816   |



| Runtime  | Qubits  | Success |
|----------|---------|---------|
| 3.95 sec | 0, 3, 2 | 0.910   |

